TECHNICAL ENGINEERING MANUAL

# Aerothon 2026: Physical UAV Architecture, Code Flow Diagrams & Deployment Manual

Complete Systems Integration Reference for Holybro X500, Cube Orange+, Raspberry Pi 5 & SIYI Camera

| AIRFRAME: | FLIGHT CONTROLLER: | COMPANION COMPUTER: | VISION SENSOR: |
|---|---|---|---|
| Holybro X500 Multirotor | Cube Orange+ (ArduPilot 4.5) | Raspberry Pi 5 (8GB) | SIYI A8 mini / ZR10 (-90° Nadir) |

**CRITICAL SAFETY MANDATE (BENCHTOP TESTING):** Always **remove all 4 propellers** before powering the multirotor on the bench. Complete visual recognition, MAVLink telemetry, and active mode switching must be verified with zero propeller hazard.

## 1. Physical Hardware Architecture & Interconnections

Autonomous visual servoing on a multirotor requires electrical isolation between motor power buses and digital computing rails to eliminate brownouts, EMF noise spikes, and ground loops during high-current flight maneuvers.

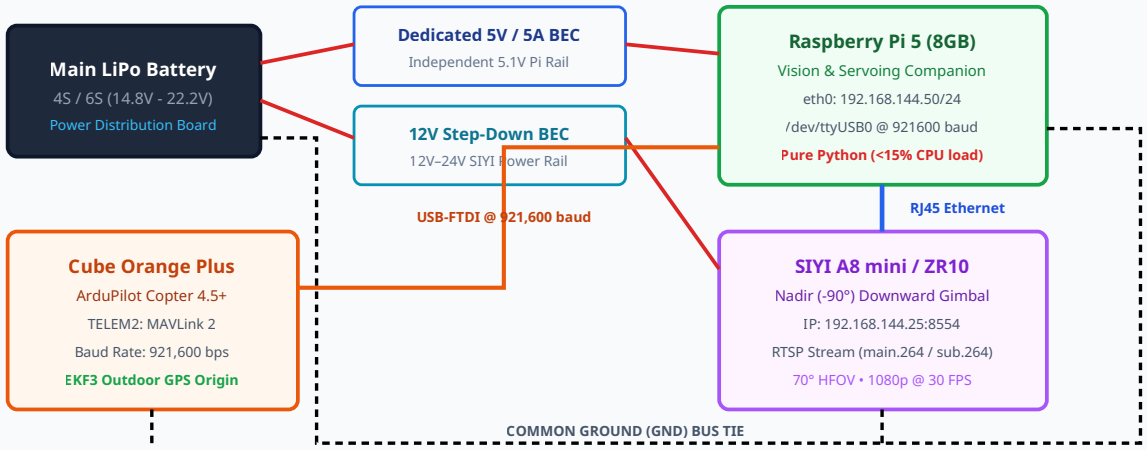



*Figure 1: Full System Electrical Schematic, Power Distribution & Inter-Process Communication Topology.*

**POWER RULE 1: NEVER POWER PI 5 FROM PIXHAWK / CUBE:** The Cube internal 5V rail is strictly rated for 1A max (for internal IMUs, barometers, and RC receivers). The Pi 5 draws up to 5A under high CPU load. It must be powered exclusively from the dedicated 5V/5A BEC wired to the main battery.

## 2. Hardware Subsystem Wiring Specifications & Parameters

| Interface | Source (Pin / Port) | Destination (Pin / Port) | Electrical Protocol | Technical Function & Rationale |
|---|---|---|---|---|
| **Pi 5 Dedicated Power** | 5V / 5A Step-Down BEC | Pi 5 USB-C or GPIO Pin 2,4 (+5V), Pin 6 (GND) | 5.1V DC steady (5A max) | Supplies stable power. Eliminates low-voltage warnings and CPU throttling under heavy OpenCV load. |
| **SIYI Camera Power** | 12V Step-Down BEC | SIYI 6-Pin Power Cable (VCC & GND) | 12V–24V DC | Powers 3-axis brushless gimbal motors and internal H.264/H.265 RTSP hardware encoder. |
| **Vision Ethernet Link** | SIYI Ethernet Port (Pins 1,2,3,6) | Pi 5 RJ45 Ethernet Port ( `eth0` ) | 100BASE-TX / RTSP H.264 | Transmits real-time 1080p video feed at `rtsp://192.168.144.25:8554/main.264` . |
| **MAVLink Telemetry Link** | Cube Orange+ `TELEM2` Port (JST-GH) | Pi 5 USB Port (USB-to-FTDI adapter) | UART Serial @ 921,600 baud, 8N1 | Bi-directional MAVLink 2: 20 Hz Local NED setpoints, attitude, GPS health, and flight mode commands. |
| **Common Ground Tie** | Main Battery Ground Pad | Cube GND + Pi 5 GND + SIYI GND | Reference Ground (0V) | Establishes a common potential reference, eliminating serial bit flipping and Ethernet packet drops. |

### Cube Orange+ Parameters (Mission Planner)

| Parameter | Value | Function |
|---|---|---|
| `SERIAL2_PROTOCOL` | `2` | MAVLink 2 protocol on TELEM2. |
| `SERIAL2_BAUD` | `921` | 921,600 baud rate (zero latency). |
| `ARMING_CHECK` | `1` | Strict pre-arm sensor health checks. |
| `FS_GCS_ENABLE` | `1` | RTL failsafe if companion link dies. |
| `FENCE_ENABLE` | `1` | Geofence containment boundary. |
| `FENCE_RADIUS` | `35.0` | Caps flight radius to 35m from pad. |
| `WPNAV_SPEED` | `150.0` | 1.5 m/s survey cruise velocity. |
| `MNT1_TYPE` | `4 / 0` | Gimbal locked nadir (-90° pitch). |

### Safety Pilot RC Manual Overrides

The safety pilot retains instantaneous override capability via transmitter switches:

- **Flight Mode Switch (Channel 5 - 3 Position):**
  - **Pos 1 (High):** `ALT_HOLD` (Full manual throttle & attitude).
  - **Pos 2 (Mid):** `LOITER` (Rock-solid GPS position hover).
  - **Pos 3 (Low):** `GUIDED` (Autonomous script control).
- **Emergency BRAKE Switch (Channel 7 or 8):**
  - Dedicated toggle switch assigned to `BRAKE` mode. Flipping this switch instantaneously halts the drone in place, immediately overriding companion automation.

### Multi-Threaded Concurrency Model on Raspberry Pi 5

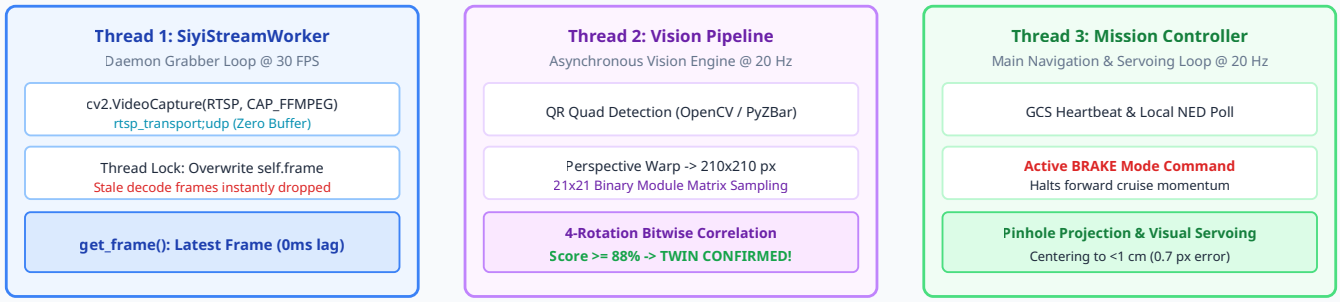



*Figure 2: Three-Thread Concurrency Architecture Eliminating RTSP Frame Queueing & Latency.*

## 3. Code Flow Diagrams: Mission FSM & Visual Servoing Mathematics

### Flow Diagram 1: Complete 6-Stage Autonomous Mission Finite State Machine




*Figure 3: Complete 6-Stage Autonomous Finite State Machine.*

### Flow Diagram 2: Pinhole Optical Projection & Closed-Loop Servoing Pipeline

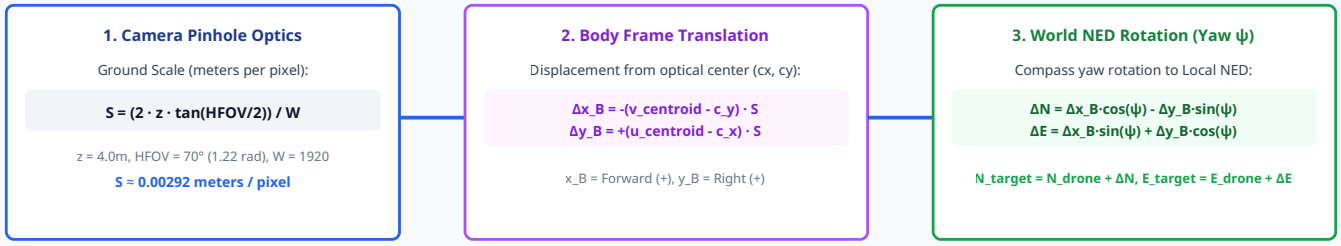



*Figure 4: Centroid Pixel Error Reprojection into Absolute Local NED Setpoint Coordinates.*

### 4-Rotation Invariant Module Grid Correlation Formula

Candidate QR modules are extracted as a $21 \times 21$ boolean module matrix $M_{\text{cand}}$. The similarity score is computed across all four discrete rotations:

```
Score = max_{k ∈ {0°, 90°, 180°, 270°}} [ (1 / 441) * ∑_{i=0}^{20} ∑_{j=0}^{20} (M_ref[i,j] == rot90(M_cand, k)[i,j]) ]
```

When `Score >= 0.88` (or decoded payload matches `AEROTHON_QR_02`), the twin candidate is confirmed, instantly initiating ArduPilot `BRAKE` mode.

## 4. Detailed Connector Pinout & Wiring Specifications

### Cable 1: Cube TELEM2 to Pi 5 USB-FTDI Pinout

| Cube TELEM2 Pin | FTDI Wire | Signal Name | Wiring Direction |
|---|---|---|---|
| Pin 1 (Red) | NC (Cut) | +5V (DO NOT CONNECT!) | Cube 5V internal rail isolated |
| Pin 2 (Black) | TXD Wire | Cube TX -> FTDI RX | Data from Autopilot to Pi |
| Pin 3 (Black) | RXD Wire | Cube RX <- FTDI TX | Data from Pi to Autopilot |
| Pin 4 (Black) | NC | CTS (Clear to Send) | Not connected (Flow control off) |
| Pin 5 (Black) | NC | RTS (Request to Send) | Not connected (Flow control off) |
| Pin 6 (Black) | GND Wire | Common Ground | Tied to Common Ground Bus |

### Cable 2 & 3: SIYI RJ45 & Dedicated Pi 5 BEC

- **SIYI Ethernet RJ45:** Connect the 8-pin aviation connector from the SIYI gimbal base directly to the Pi 5's onboard RJ45 Gigabit Ethernet port. (Pre-configured static profile `siyi-eth` @ `192.168.144.50/24`).
- **Pi 5 Dedicated BEC:** Connect a 5V/5A step-down BEC directly from the main LiPo. Connect its 5.1V output to Pi 5 USB-C or GPIO Pin 2 (+5V) and Pin 6 (GND).
- **Ground Tie:** Ensure a 22AWG ground wire bridges the battery ground, Pi 5 ground, Cube ground, and SIYI ground.

## 5. Phased Testing Protocol (From Benchtop to Full Flight)

### 1 Phase 1: SIYI Camera Stream Verification

Verifies physical Ethernet link and zero-latency stream:

```
python3 scripts/verify_siyi_stream.py
```

- Pings `192.168.144.25` with <2ms latency.
- Verifies 1080p @ 30 FPS stream with sub-frame lag.
- Check: Wave hand under lens -> movement is instantaneous.

### 2 Phase 2: Cube MAVLink Telemetry Audit

Verifies autopilot serial link at 921,600 baud:

```
python3 scripts/test_mavlink_cube.py --auto --test-modes
```

- Verifies MAVLink 2 heartbeat and battery telemetry.
- Confirm RC switches: Ch 5 (Mode) & Ch 7/8 (BRAKE).
- Verifies `COMMAND_ACK: ACCEPTED` on mode change.

### 3 Phase 3: Stage 1 Benchtop Test (PROPS OFF! ⚠️)

Tests full memory caching, distractor rejection, & BRAKE:

```
python3 scripts/stage1_bench_test.py
```

- **Step A:** Hold reference QR under lens -> `[SUCCESS] REFERENCE QR MEMORIZED`.
- **Step B:** Present distractor QR -> HUD shows ORANGE box: `Similarity: 73% [NO MATCH]`.
- **Step C:** Present twin match -> HUD shows GREEN box: `100% MATCH`, Cube switches to `BRAKE`!

### 4 Phase 4: Companion Single-Waypoint Flight

Tests manual flight stability and single translation:

```
python3 scripts/fly_to.py --target-n 5.0 --target-e 0.0 --target-d -4.0 --url /dev/ttyUS
```

- Hover manually in `LOITER` at 4m altitude.
- Flip RC switch to `GUIDED` -> Drone executes 5m translation.
- Flip switch back to `LOITER` -> verifies manual takeover works instantly.

### 5 Phase 5: Stage 4 Full Autonomous Mission Execution

Place reference QR tile on takeoff pad, candidate QR tiles in search field, ensure geofence is active, and launch:

```
python3 scripts/lawnmower_qr_mission_real.py --url /dev/ttyUSB0:921600 --alt 4.0 --speed 1.5 --step 4.5 --align-duration 1
```

**Expected Autonomous Sequence:** Takeoff to 4.0m • Memorize reference QR • Transition forward • Serpentine search • Detect twin • Instant BRAKE • PBVS centering to <1 cm • 15-second steady alignment hover • Land/RTL.